

# MUHAMMAD HEDIA ABD-ELHAMED

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## SUMMARY

Backend Developer and recent Computer Science graduate, focused on Java backend engineering with Spring Boot and PostgreSQL. Built role-aware platforms, event-driven services, financial reconciliation systems, and reproducible database performance experiments across Java 21 and Java 25. Project work spans Docker, CI/CD, Kubernetes, and AWS serverless workflows validated locally with LocalStack.

## SKILLS

- **Languages:** Java, Python, C++, C#, JavaScript, SQL
- **Backend:** Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs
- **Data:** PostgreSQL, Oracle SQL/PLSQL, MongoDB, Flyway
- **Systems & Cloud:** Linux, Apache Kafka, Event-Driven Architecture, Kubernetes, AWS (SQS, Lambda, DynamoDB, S3, IAM; LocalStack validated), Terraform
- **Testing & Delivery:** JUnit 5, Mockito, Testcontainers, OpenAPI/Swagger, Docker, Docker Compose, Git, GitHub Actions, Maven

## EDUCATION

### Cairo University - Faculty of Science

B.Sc. in Computer Science | Oct 2021 - Jan 2026 | GPA: 3.289 out of 5 | Very Good with Honors

## PROJECTS

### MerHouse - B2B Fulfillment Coordination Platform

Java 21 | Spring Boot | Spring Security | Spring Data JPA | PostgreSQL | React/TypeScript | Docker | GitHub Actions

- Built a role-aware B2B fulfillment platform for merchants, warehouse providers, and platform operators, covering inventory, inbound stock, orders, allocation, fulfillment, exceptions, shipments, notifications, and service accountability.
- Implemented tenant-aware Spring Boot REST APIs with Spring Security and PostgreSQL, including transactional outbox, inventory locking, partial allocation/backorders, shipment state transitions, and OpenAPI.
- Delivered React/TypeScript and Capacitor Android clients with automated tests. Packaged with Docker Compose and automated CI through GitHub Actions.

### EventFlow - Distributed Event-Driven Backend

Java 25 | Spring Boot 3.5 | Apache Kafka | Kubernetes | AWS | Terraform | LocalStack | GitHub Actions

- Built two independently deployable Spring Boot services using Kafka, at-least-once processing, event-ID idempotency, retries, and dead-letter handling; deployed with Strimzi on Kubernetes and verified in-cluster smoke tests.
- Implemented an AWS serverless variant using SQS, Lambda (Java 21), DynamoDB, S3, IAM, and Terraform. Validated locally with LocalStack, including IAM enforcement and durable idempotency; no production AWS deployment is claimed.

### QueryForge - PostgreSQL Performance Engineering Service

Java 25 | Spring Boot 3.5.5 | PostgreSQL 17 | Docker | k6

- Built reproducible k6 benchmarks for a deterministic 1M-row PostgreSQL catalog; reduced filtered-search p95 from 8.18 s to 130 ms (~63x) at 10 req/s with a workload-shaped composite index.
- Compared OFFSET and keyset pagination, index ordering, and workloads up to 200 concurrent users; identified the 10-connection pool as the next measured bottleneck using EXPLAIN (ANALYZE, BUFFERS).

### Reconcile - Payment & Settlement Reconciliation Engine

Java 25 | Spring Boot | Spring Security | Spring Data JPA | PostgreSQL 18 | Flyway | Docker | GitHub Actions

- Built a double-entry payments backend where the database enforces the financial invariants - 23 tables, 36 indexes, 10 triggers - across a create-authorize-capture-settle-refund lifecycle, payouts, and an append-only audit trail reconstructable from the database alone.
- Ingested signed provider webhooks over the raw bytes with replay protection, proved idempotency under 32 concurrent retries, and reconciled ledger-derived expectations against settlement records deterministically across 50 subjects and all nine outcomes, byte for byte; 354 tests (353 passed, 1 deliberately skipped, 0 failed). Validated prototype: not production-ready, not PCI compliant, no real payment processing.

## ACHIEVEMENTS

Ranked #1 nationally in Computer Science and #7 nationally in Mathematics across Faculties of Science, Egypt (2026 graduates) | #2 in Mathematics at Cairo University

## LANGUAGES

Arabic - Native | English - C1-level proficiency